

ADA: Demographic Methods

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Week 6: Migration

Overview

- ▶ Migration intro
 - ▶ Data sources
 - ▶ Adding into a cohort component projection framework
- ▶ Migration models
 - ▶ Two population Leslie Matrix
 - ▶ Gravity and log-linear models
 - ▶ Model age patterns

Migration

Back to the demographic accounting identity

Recall from week 1:

$$P(t+1) = P(t) + B[t, t+1] - D[t, t+1] + I[t, t+1] - O[t, t+1]$$

or

$$P(t+1) = P(t) + B[t, t+1] - D[t, t+1] + N[t, t+1]$$

where $N[t, t+1]$ is net migration.

- ▶ migration is the third component of population change
- ▶ generally less contribution to population change, but becoming more important
- ▶ we've largely ignored it until now, focusing on natural increase/decrease
- ▶ adding migration messes up our idea of generational renewal, etc

Why is migration important?

From a demographic perspective, migration changes the age and sex structure of a population

- ▶ can offset natural decrease
- ▶ reduce dependency ratios (economic implications)

Old age dependency ratio:

$$\frac{\text{Population aged 65+}}{\text{Population aged 15-64}}$$

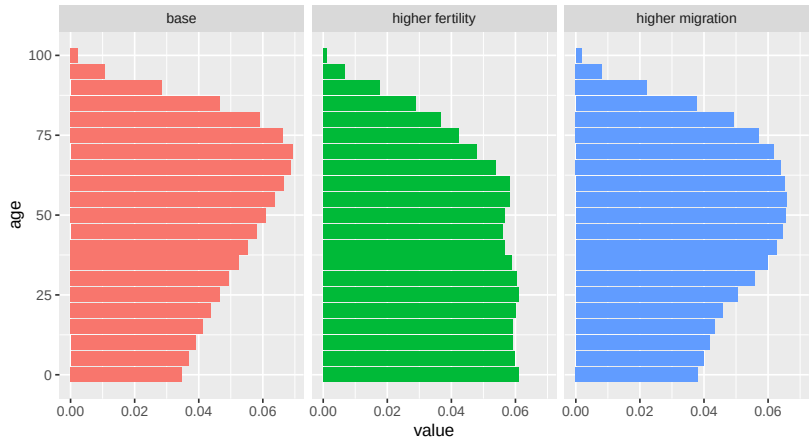
Child dependency ratio:

$$\frac{\text{Population aged 0-14}}{\text{Population aged 15-64}}$$

Is increasing migration as effective as increasing fertility?

Migration versus fertility effects on population age structure

Canada's population age structure, 2120
Baseline, replacement fertility, and migration increased by 25%



Issues

Definition issues:

- ▶ not biological
- ▶ geographical element
- ▶ depends on intent or subsequent behavior
- ▶ migration versus mobility (temporal component)
- ▶ contextually dependent (international versus internal)

Measurement issues:

- ▶ separate from vital registration
- ▶ immigration easier than emigration
- ▶ illegality issues

Measurement

$$P(t + 1) = P(t) + B[t, t + 1] - D[t, t + 1] + N[t, t + 1]$$

If we knew population size (e.g. from census), births and deaths completely accurately, net migration is just the residual

But in reality, we are usually dealing with

$$P(t + 1) = P(t) + B[t, t + 1] - D[t, t + 1] + N[t, t + 1] + \text{error}$$

Measures of migration

Stock: Population that are living in geographic area of interest (state, country) who were not born in that area. Gives idea of overall magnitude but no information about recent changes.

Flows: The number of people moving into (or out of) a geographic area over a time period of interest. Changes in stocks gives some information about flows.

Migration rates: What is the denominator? For emigration it is as before, but for immigration it is not usually the population at risk.

Data sources

Traditional data come from censuses or surveys

- ▶ Inferred from population change
- ▶ Birthplace
- ▶ Place of previous residence
- ▶ Duration of current residence

Example: ACS

15 a. Did this person live in this house or apartment 1 year ago?

☐ Person is under 1 year old → *SKIP to question 16*

☐ Yes, this house → *SKIP to question 16*

☐ No, outside the United States and Puerto Rico – *Print name of foreign country, or U.S. Virgin Islands, Guam, etc., below; then SKIP to question 16*

☐ No, different house in the United States or Puerto Rico

b. Where did this person live 1 year ago?

Address (Number and street name)

Name of city, town, or post office

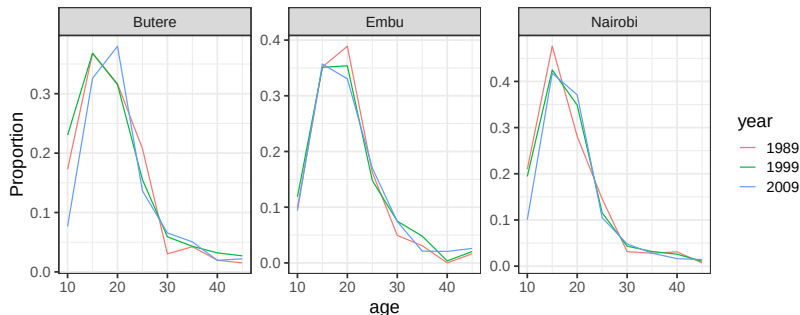
Name of U.S. county or municipio in Puerto Rico

Name of U.S. state or Puerto Rico

ZIP Code

Example: Kenya

From the Census we know district of residence one year ago. Can work out net migration by looking at the difference between in and out migrants. Proportion of migrants by age:



Adding migration into a cohort component
projection framework

Adding migration into a CCP

Remember from last week, we have the Leslie Matrix population projection set-up:

$$\begin{bmatrix} \text{kids} & \text{kids} & \text{kids} & \dots & \text{kids} \\ \frac{{}_5L_{10}}{{}_5L_0} & 0 & 0 & \dots & 0 \\ 0 & \frac{{}_5L_{15}}{{}_5L_{10}} & 0 & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & 0 \end{bmatrix}$$

where the kids entries are:

$${}_nL_0 \cdot \frac{1}{2} \left({}_nF_x + {}_nF_{x+n} \cdot \frac{{}_nL_{x+n}}{{}_nL_x} \right) \cdot f_{fab}$$

Adding migration into a CCP

Call this A . Multiply population vector $K(t)$:

$$K(t) = \begin{bmatrix} {}_5K_0 \\ {}_5K_5 \\ {}_5K_{10} \\ \dots \\ {}_{\infty}K_{\omega} \end{bmatrix}$$

by A to get $K(t+n)$.

$$K(t+n) = AK(t)$$

Adding migration into a CCP

Think about the period t to $t + n$ (e.g. often $n = 5$ so this is a five-year period).

- ▶ if all migrants arrived at the end of the period, then we would have

$$K(t+5) = \begin{bmatrix} \text{kids} & \text{kids} & \text{kids} & \dots & \text{kids} \\ \frac{{}_5L_{10}}{{}_5L_0} & 0 & 0 & \dots & 0 \\ 0 & \frac{{}_5L_{15}}{{}_5L_{10}} & 0 & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & 0 \end{bmatrix} \times K(t) + \begin{bmatrix} {}_5K_0(t) {}_5i_0 \\ {}_5K_5(t) {}_5i_5 \\ {}_5K_{10}(t) {}_5i_{10} \\ \dots \\ {}_\infty K_\omega(t) {}_\infty i_\omega \end{bmatrix}$$

where ${}_n i_x$ is the number of net migrants expressed as a proportion of population size, so that ${}_n K_x \cdot {}_n i_x = {}_n l_x$ is the number of net migrants.

Adding migration CCP

Perhaps a slightly better approximation is assuming that half the migrants arrive at the start of the period and half arrive at the end. Then we have:

$$\begin{aligned}
 K(t+5) = & \begin{bmatrix} \text{kids} & \text{kids} & \text{kids} & \dots & \text{kids} \\ \frac{{}_5L_{10}}{{}_5L_0} & 0 & 0 & \dots & 0 \\ 0 & \frac{{}_5L_{15}}{{}_5L_{10}} & 0 & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & 0 \end{bmatrix} \times \begin{bmatrix} {}_5K_0(t) + {}_5K_0(t){}_5i_0/2 \\ {}_5K_5(t) + {}_5K_5(t){}_5i_5/2 \\ {}_5K_{10}(t) + {}_5K_{10}(t){}_5i_{10}/2 \\ \dots \\ {}_\infty K_\omega(t) + {}_\infty K_\omega(t){}_\infty i_\omega/2 \end{bmatrix} \\
 & + \begin{bmatrix} {}_5K_0(t){}_5i_0/2 \\ {}_5K_5(t){}_5i_5/2 \\ {}_5K_{10}(t){}_5i_{10}/2 \\ \dots \\ {}_\infty K_\omega(t){}_\infty i_\omega/2 \end{bmatrix}
 \end{aligned}$$

So half the migrants come in at the start and have time to die/give birth, and half come in at the end of the period.

Adding migrants to CCP

We are assuming that as soon as migrants arrive they experience the same mortality and fertility conditions as the native population.

- ▶ In practice, depending on the size of the population, this probably doesn't matter much
- ▶ How realistic is this assumption in practice?
 - ▶ fertility: usually depends on age at migration, education etc (Adsera and Ferrer (2011) illustrate this for Canada)
 - ▶ mortality: depends on migrant group, often observed to be higher in high-skilled migrant groups ('healthy migrant effect')

Migration models

Overview

- ▶ The focus here is on aggregate models, not individual models
 - ▶ But there is an increasing use of agent based models in migration estimation and projection
- ▶ Demographers are mostly interested in two dimensions:
 1. net flows into areas (including pairwise relationships between areas)
 2. age patterns of migration

Two population Leslie Matrix

- ▶ Interested in capturing movements between two areas (e.g. urban-rural movements)
- ▶ Split Leslie Matrix into distinct submatrices based on four populations: urban natives, rural natives, urban migrants and rural migrants.

$$\begin{bmatrix} M_{\text{city native}}^{\text{city native}} & M_{\text{city native}}^{\text{city migrant}} & 0 & 0 \\ 0 & M_{\text{city migrant}}^{\text{city native}} & M_{\text{city migrant}}^{\text{village native}} & M_{\text{city migrant}}^{\text{village migrant}} \\ 0 & 0 & M_{\text{village native}}^{\text{city migrant}} & M_{\text{village native}}^{\text{village migrant}} \\ M_{\text{village migrant}}^{\text{city native}} & M_{\text{village migrant}}^{\text{city migrant}} & 0 & M_{\text{village migrant}}^{\text{village migrant}} \end{bmatrix}$$

- ▶ migrant to migrant transitions represent return migration (could simplify by assuming no return migration)
- ▶ migrant to native transitions represent births of migrants

Two population Leslie Matrix

$$\begin{bmatrix} M_{\text{city native}}^{\text{city native}} & M_{\text{city migrant}}^{\text{city native}} & 0 & 0 \\ 0 & M_{\text{city migrant}}^{\text{city migrant}} & M_{\text{city migrant}}^{\text{village native}} & M_{\text{city migrant}}^{\text{village migrant}} \\ 0 & 0 & M_{\text{village native}}^{\text{village native}} & M_{\text{village native}}^{\text{village migrant}} \\ M_{\text{village migrant}}^{\text{city native}} & M_{\text{village migrant}}^{\text{city migrant}} & 0 & M_{\text{village migrant}}^{\text{village migrant}} \end{bmatrix}$$

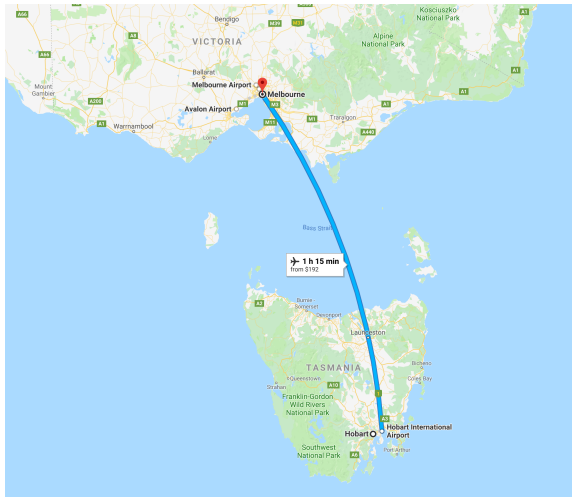
The elements of each M are: Fertility rates for the four populations; Survival probabilities for the four populations; Probabilities of migration.

Example



Hobart

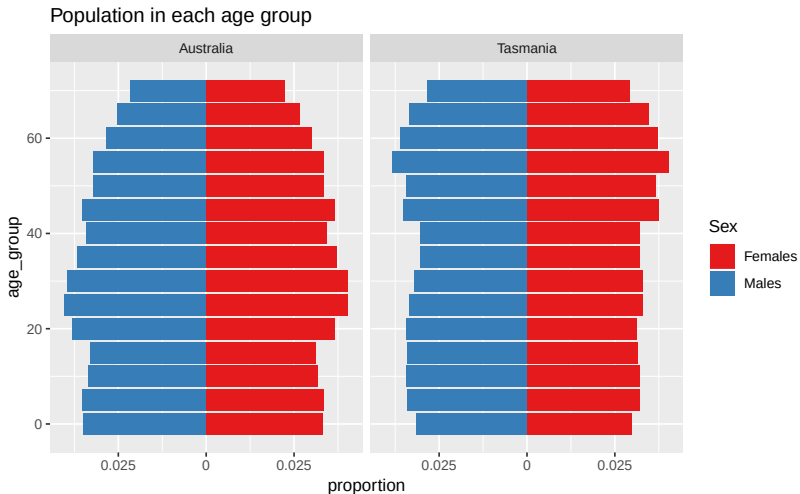




Melbourne



Population age pyramids

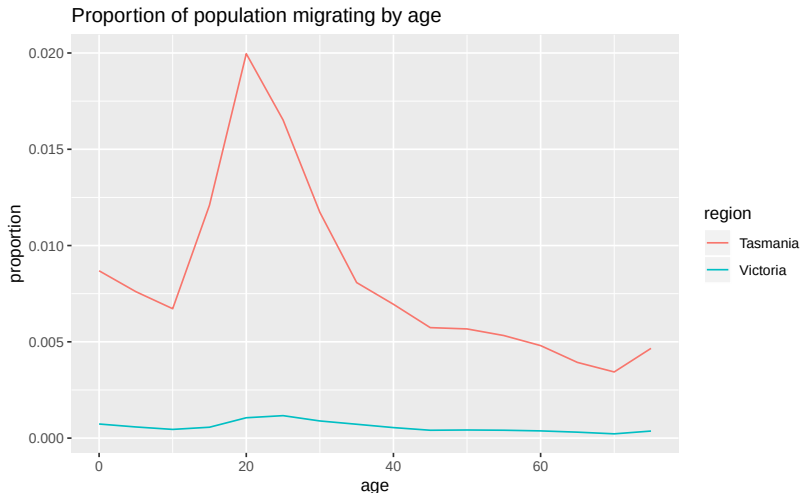


Set up two-population Leslie Matrix

$$\begin{bmatrix} M_{\text{city native}}^{[\text{city native}]} & M_{\text{city migrant}}^{[\text{city migrant}]} & 0 & 0 \\ 0 & M_{\text{city migrant}}^{[\text{city migrant}]} & M_{\text{city migrant}}^{[\text{village native}]} & M_{\text{city migrant}}^{[\text{village migrant}]} \\ 0 & 0 & M_{\text{village native}}^{[\text{village native}]} & M_{\text{village native}}^{[\text{village migrant}]} \\ M_{\text{village migrant}}^{[\text{city native}]} & M_{\text{village migrant}}^{[\text{city migrant}]} & 0 & M_{\text{village migrant}}^{[\text{village migrant}]} \end{bmatrix}$$

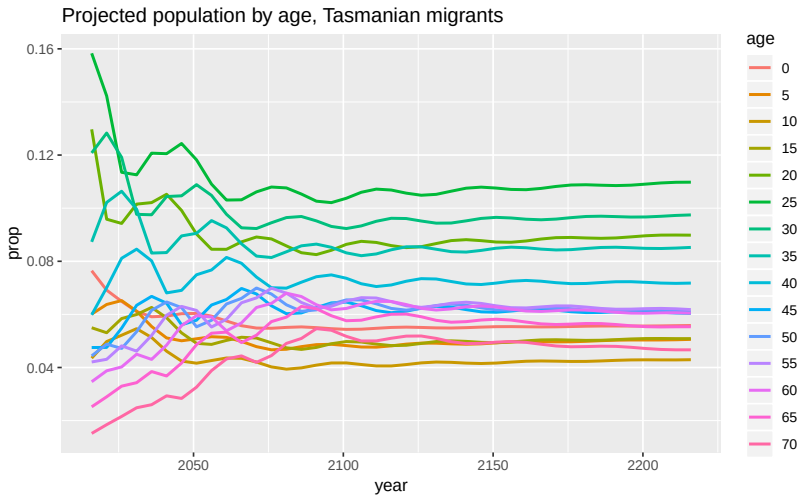
- ▶ Tasmania is 'village'
- ▶ Victoria is 'city'
- ▶ Note: assume that migrants are always migrants, but they give birth to natives

Migration by age

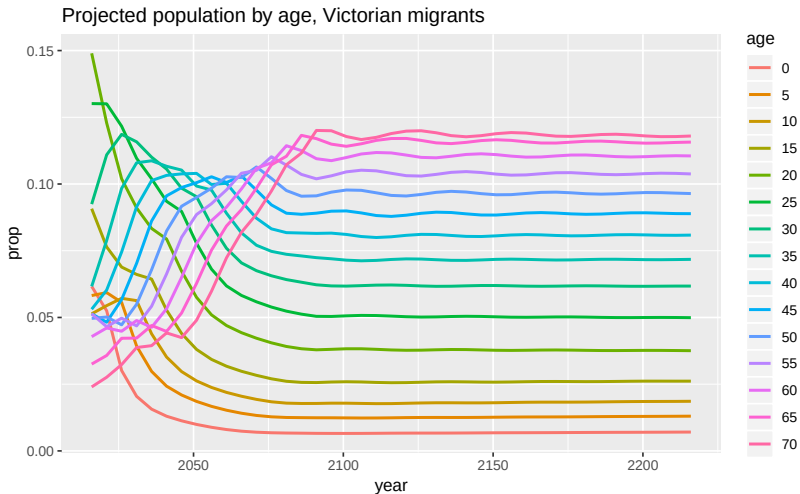


- ▶ majority (95%) of Tasmanians do not return after migrating
- ▶ majority (80%) of Victorians return after migrating

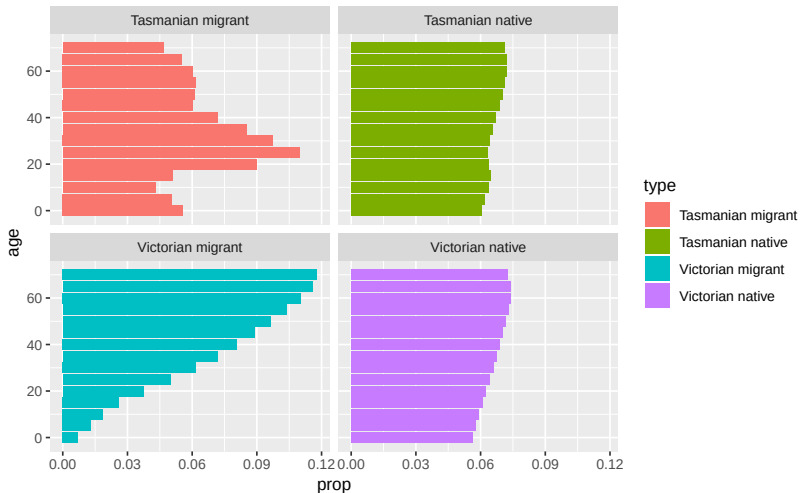
Projections of migrant populations



Projections of migrant populations



Stable populations



Gravity model

Resembles Newton's law of gravity. The idea that spatial interaction is related to stocks and inversely related to distance.

$$M_{ij} = G \frac{K_i^\alpha K_j^\beta}{D_{ij}^\gamma}$$

- ▶ G is some proportionality constant
- ▶ K_i is population of area i
- ▶ K_j is population of area j
- ▶ α is strength of 'push' factors
- ▶ β is strength of 'pull' factors
- ▶ D_{ij} is distance

Gravity model

On the log scale

$$\log M_{ij} = \log G + \alpha \log K_i + \beta \log K_j - \gamma \log D_{ij}$$

- ▶ Easy to fit
- ▶ But:
 - ▶ predicts symmetric flows between i and j
 - ▶ doesn't include push/pull factors other than population and distance

A more general representation

$$\log M_{ij} = \mu + \alpha_i + \beta_j + \gamma_{i,j}$$

for $i \neq j$.

- ▶ Push/pull factors need not just be population size
- ▶ Interaction effect need not be a linear function of distance

Note that this is a saturated model, not really a model, but a way of representing the data.

To make identifiable, set $\alpha_0 = \beta_0 = \gamma_{1j} = \gamma_{i1} = 0$.

Log-linear models

Quasi-independent generally fits reasonably well to large populations:

$$\log M_{ij} = \mu + \alpha_i + \beta_j$$

for $i \neq j$. 'Quasi' because we are not including the diagonal terms in the model.

Could also put some structure on the interaction term, for example

$$\log M_{ij} = \mu + \alpha_i + \beta_j + \gamma d_{ij}$$

$d_{ij} = 1$ when areas are adjacent and zero otherwise.

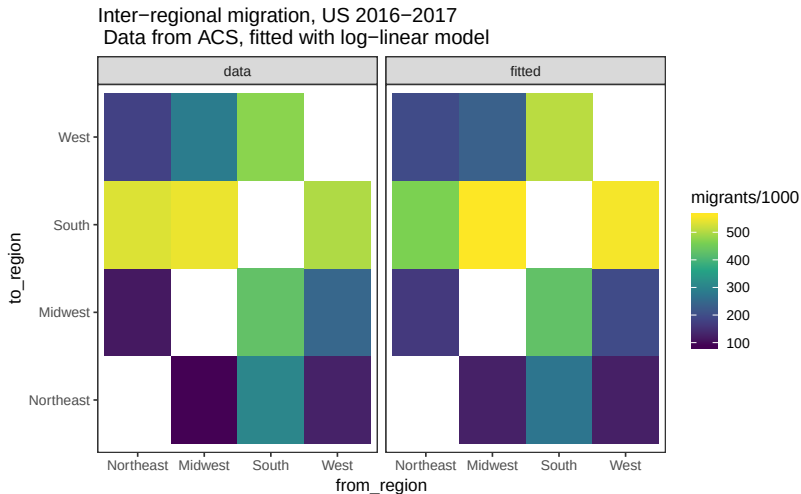
Example

US inter-regional movements, 2016-2017. Looking at migrants only, so diagonals are 0.

	2017				
2016	Northeast	Midwest	South	West	Total
Northeast	0	116096	529678	179832	825606
Midwest	92079	0	540847	284665	917591
South	304617	427617	0	471788	1204022
West	128870	246592	496148	0	871610
Total	525566	790305	1566673	936285	3818829

Fit

using `glm` in R:



Using log-linear models with missing data

In some cases, we may not have information on all cells.

Example: migration between countries (e.g. Europe)

- ▶ probably know marginal totals (i.e. number of migrants in each countries) M_{i+} and M_{+j}
- ▶ may only have information on source / destination breakdown for some countries

As long as we know marginals and have some information on the cells, we can use a log-linear model to fill in estimates (Raymer 2007).

Model age patterns of migration

Migration often occurs in conjunction with some transition in the life course:

- ▶ Adult migration peaks at young adult ages (work, education)
- ▶ Second peak around retirement
- ▶ Child peak associated with parents' migration

Model age patterns of migration

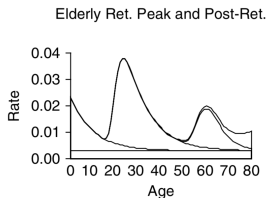
Multi-exponential model of migration (Rogers and Castro, 1981).
Migration rate $m(x)$ is modeled as:

$$\begin{aligned} m(x) = & a_1 \exp(-\alpha_1 x) + a_2 \exp(-\alpha_2(x - \mu_2) - \exp[\gamma_2(x - \mu_2)]) \\ & + a_3 \exp(-\alpha_3(x - \mu_3) - \exp[\gamma_3(x - \mu_3)]) + c \end{aligned}$$



Model age patterns of migration

$$m(x) = a_1 \exp(-\alpha_1 x) + a_2 \exp(-\alpha_2(x - \mu_2) - \exp[\gamma_2(x - \mu_2)]) \\ + a_3 \exp(-\alpha_3(x - \mu_3) - \exp[\gamma_3(x - \mu_3)]) + c$$



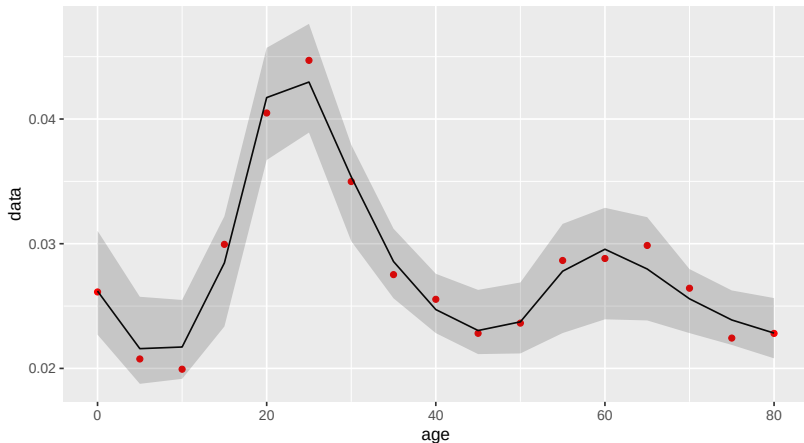
- ▶ α 's and γ 's are rates of descent and ascent
- ▶ μ 's are peak ages
- ▶ all others are intensity parameters

Example

We can fit this horrible looking model in a Bayesian framework using Stan.

For Florida in 2017: $\mu_1 = 24.5$, $\mu_2 = 60$.

Florida migration age schedule, 2017
Data and Rogers-Castro fit



More when we get to Bayes

VOLUME 49 - ARTICLE 42 | PAGES 1201–1228

Bayesian implementation of Rogers–Castro model migration schedules: An alternative technique for parameter estimation

BY Jessie Yeung, Monica Alexander, Tim Riffe

Social media data and migration

Migration data

- ▶ Often migration information available from traditional/official sources is sparse or delayed in release
- ▶ Lots of examples of researchers using interesting sources of information to try and measure migration and mobility
 - ▶ cell phone data (mobility, short-term)
 - ▶ school enrollments
 - ▶ tax filings
 - ▶ social media (Twitter, Facebook/Instagram, LinkedIn)

Social media and migration

The screenshot displays the Facebook Ads Manager interface. On the left, a sidebar contains navigation links: Campaign, Ad Account, Ad Set (selected), and Ad. The main area is titled 'Ad Set Name' and shows the configuration for an ad set named '30-39'. The 'Locations' section is set to 'Everyone in this location' and shows a map of the United States with 'California' selected, indicated by a red arrow. The 'Age' range is set to '30 - 39', also indicated by a red arrow. The 'Gender' is set to 'Women'. The 'Detailed Targeting' section shows 'Behaviors > Expats' with 'Expats (Australia)' selected, indicated by a red arrow. On the right, a sidebar provides additional information: 'Create Multiple Ad Sets in One Step', 'Audience Size' (a gauge showing 'Specific' and 'Broad' ranges with a needle pointing towards 'Broad', circled in red), 'Potential Reach: 4,600 people', and 'Estimated Daily Results' (Reach: 650 - 1,300).

Facebook Ads Manager interface showing Ad Set configuration for '30-39'.

Locations: Everyone in this location. United States. **California** (highlighted with a red arrow).

Age: 30 - 39 (highlighted with a red arrow).

Gender: All, Men, **Women** (highlighted with a red arrow).

Detailed Targeting: INCLUDE people who match at least ONE of the following. Behaviors > Expats. **Expats (Australia)** (highlighted with a red arrow).

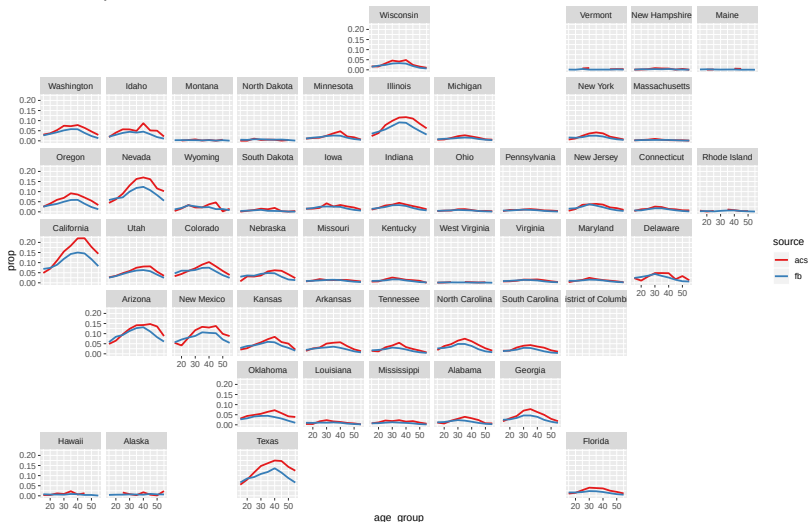
Audience Size: Your audience selection is fairly broad. Potential Reach: 4,600 people (circled in red).

Estimated Daily Results: Reach 650 - 1,300.

Example: Mexican migrant age distributions in the USA

Facebook versus ACS

FB v ACS by state



Bias is substantial, but can be modeled

- ▶ Adjust Facebook data based on gold standard
- ▶ Build time series model incorporating historical trends and new Facebook data
- ▶ Projections are informed by both

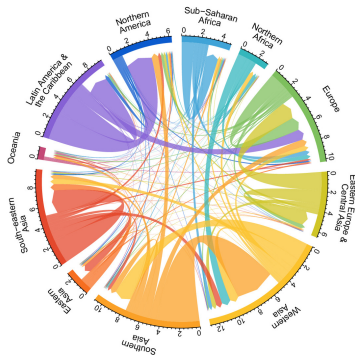
An attempt from me et al.:

<https://link.springer.com/article/10.1007/s11113-020-09599-3>

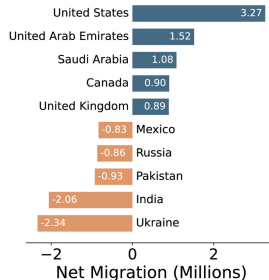
Global bi-lateral flows

from Chi et al. 2025

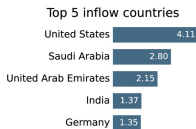
A



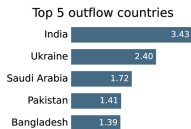
B



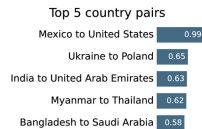
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D



E



Example: out-migration after a natural disaster

The Impact of Hurricane Maria on Out-migration from Puerto Rico: Evidence from Facebook Data

MONICA ALEXANDER 

KIVAN POLIMIS

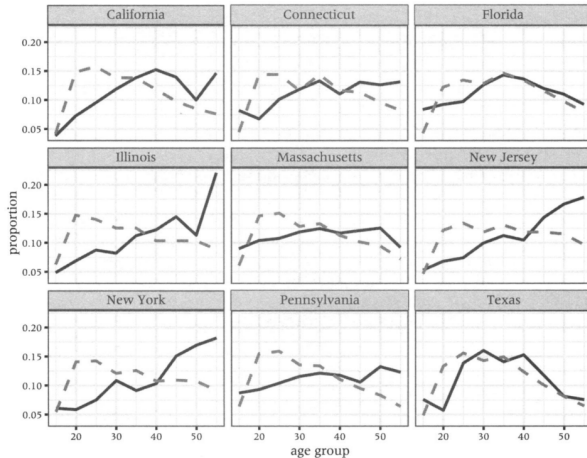
EMILIO ZAGHENI

Hurricane Maria

- ▶ Hit multiple Caribbean islands in September 2017
- ▶ Huge out-migration from Puerto Rico to the United States
- ▶ But no official data sources available in a timely manner
- ▶ We had been collecting Facebook data for another project and realised we had a measure of Puerto Ricans in continental US before and after the hurricane

Biases

FIGURE 1 Age distribution of Puerto Rican migrants, January 2017



NOTES: Facebook data in January 2017 (dashed line) and 2017 American Community Survey data (solid line).

Methods

Methods

To address the data challenges described above, our method of estimating the extent of out-migration from Puerto Rico involved two main steps: using Difference-in-Differences to obtain estimates of the percent change in migrants, and then applying these to ACS data to obtain population estimates by demographic subgroups.

Methods

Define π_g to be the proportional change in Puerto Rican migrants over the period October 2017 to January 2018 (i.e. Waves 4 and 5 in our data) for group of interest g . The group g might be all Puerto Rican migrants at the national level, or stratified by any age, sex and state group—for example Puerto Rican migrants aged 30–34 in Florida. We estimate π_g as

$$\hat{\pi}_g = \frac{P_g^{PR}(5) - P_g^{PR}(4)}{P_g^{PR}(4)} - \frac{P_g^C(5) - P_g^C(4)}{P_g^C(4)}$$

where $P_g^{PR}(w)$ refers to the population of Puerto Rican migrants in group g at wave w (where $w = 4$ or 5), and $P_g^C(w)$ refers to the population in the control group for the same group and wave.

Results

TABLE 2 Estimated increase in Puerto Rican migrant stocks, October 2017 to January 2018

State	% Increase (95% CI)	Population Increase
Florida	21.6 (20.9, 22.3)	65433 (63342, 67525)
New York	11.0 (10.3, 11.7)	14477 (13584, 15371)
Pennsylvania	13.4 (12.7, 14.1)	13441 (12700, 14181)
Connecticut	14.7 (12.9, 16.5)	9402 (8244, 10560)
Massachusetts	10.1 (8.82, 11.4)	8957 (7824, 10090)
Texas	10.8 (10.4, 11.2)	5678 (5452, 5904)
Ohio	12.8 (12.2, 13.4)	3274 (3125, 3424)
Illinois	9.90 (9.15, 10.6)	2641 (2441, 2841)
Georgia	13.1 (12.4, 13.8)	2606 (2470, 2742)
New Jersey	2.90 (1.56, 4.24)	2282 (1228, 3336)
California	2.40 (1.86, 2.94)	573 (444, 702)

NOTE: The 95% confidence intervals are shown in parentheses.

Results

TABLE 4 Return migration: Estimated change in Puerto Rican migrant stocks, January 2018 to March 2018

State	% Change	Population Change
Florida	-7.1 (-7.77, -6.43)	-21508 (-23526, -19490)
Massachusetts	-4.5 (-5.52, -3.48)	-3991 (-4899, -3083)
Connecticut	-3.6 (-5.04, -2.16)	-2302 (-3226, -1379)
Texas	-3.5 (-3.91, -3.09)	-1840 (-2055, -1625)
Pennsylvania	-1.4 (-2.03, -0.773)	-1404 (-2033, -776)
Ohio	-1.7 (-2.14, -1.26)	-435 (-548, -322)
New York	0.4 (-0.241, 1.04)	526 (-318, 1371)
Illinois	2.8 (2.15, 3.45)	747 (574, 920)
Georgia	6.4 (5.86, 6.94)	1273 (1165, 1381)
New Jersey	2.3 (1.15, 3.45)	1810 (908, 2712)
California	8.5 (7.97, 9.03)	2029 (1902, 2157)

NOTE: The 95% confidence intervals are shown in parentheses.